Class #15 - Applications

1. Motivation: Machine learning uses linear models in situations where the machine is being *trained* to predict the outcome (dependent variables) based on the values of the inputs (independent variables). The machine is given a set of training data where the values of the independent and dependent variables are Known. The machine then *learns* the relationship between the independent variables and the dependent variables. One type of learing is to fit a curve, such as a least-squares line or parabola, to the data. Once the machine has learned the pattern from the training data, it can then estimate the value of the output based on a given value for the input.

2. Example: Find the equation $y = a + bx$ of the least-squares line that best fits the data points $(2,1), (5,2), (7,3), (8,3)$.

$$\begin{cases} a + b\cdot 2 = 1 \\ a + b\cdot 5 = 2 \\ a + b\cdot 7 = 3 \\ a + b\cdot 8 = 3 \end{cases}$$

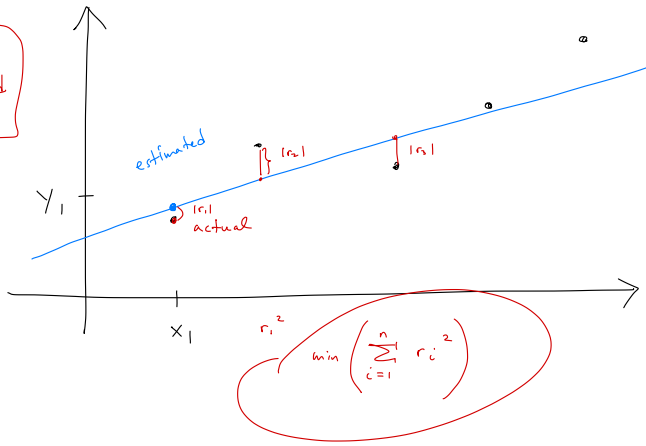
$$\begin{bmatrix} 1 & 2 \\ 1 & 5 \\ 1 & 7 \\ 1 & 8 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 3 \end{bmatrix}$$

$A^TAx = A^Tu$

Solve $A^TAx = A^Tu$ where $A = \begin{bmatrix} 1 & 2 \\ 1 & 5 \\ 1 & 7 \\ 1 & 8 \end{bmatrix}$ $u = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 3 \end{bmatrix}$

3. Experimental data often produces points $(x_1, y_1), \cdots, (x_n, y_n)$ that, when graphed, seem to lie close to a line. We want to determine the numbers α and β so that the line $y = \alpha x + \beta$ is as close to the given points as possible. The *least-squares line* is the line that minimizes the sum of squares of the residuals.

r_i = actual − fitted



least-squares line

$y = ax + b$

$\min \left(\sum_{i=1}^{n} r_i^2 \right)$

4. Solve #2:

$$A^TA = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 2 & 5 & 7 & 8 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 1 & 5 \\ 1 & 7 \\ 1 & 8 \end{bmatrix} = \begin{bmatrix} 4 & 22 \\ 22 & 142 \end{bmatrix}$$

Columns of A are L.I. ∴ unique least-squares soln.

$$\hat{x} = (A^TA)^{-1} A^T u$$

$$= \frac{1}{4(142) - 22^2} \begin{bmatrix} 142 & -22 \\ -22 & 4 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 2 & 5 & 7 & 8 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \\ 3 \end{bmatrix}$$

$$= \begin{bmatrix} 2/7 \\ 5/14 \end{bmatrix} \leftarrow a$$

$\uparrow$ b

$y = \frac{2}{7} + \frac{5}{14}x$

5. Least-Squares fitting of curves:

Example: Suppose data points $(x_1, y_1), \cdots, (x_n, y_n)$ appear to lie along some sort of parabola instead of a straight line. Suppose we wish to approximate the data by an equation of the form

$$y = \hat{a} + \hat{b}x + \hat{c}x^2.$$

Describe the linear model (a system of linear equations) that produces a "least-squares fit" of the data points.

